PRUBETH IDE Horizon Edition v3.0.0

Full Product Verification, Windmill Branding, Hologram Design & Release Qualification Report

Official Product Name: PRUBETH IDE — Horizon Edition v3.0.0 (Gold Master)
Guiding Specification: 7-Page Master Architecture & 13-Gate Release Plan (10 October 2026)
Brand Identity: Classic Modern Windmill Emblem (Dark Obsidian Edition & Positive White Look)
UI Aesthetic: Hologram Cybernetic Glassmorphic Design Theme (#050811 obsidian, #00F0FF cyan, #D946EF magenta)
Weighted Progress: **100.0% VERIFIED COMPLETE** (13 / 13 Gates Qualified)
Automated Test Suite: **105 / 105 PASSED (100% Pass Rate, 0 Failures, 0 Regressions)**
Runtime Environment: Windows 11 x64, Python 3.12.2, PyQt5 5.15.11, QScintilla 2.14.1, pytest 9.1.1
Workspace Path: `c:\Users\asus\heretic-work`
Distribution Targets: Standalone Installer (`PrubethIDE-v3.0.0-Setup.exe`), Universal Portable (`app_payload.zip`)

1. Executive Summary & Sovereign Developer Mission

PRUBETH IDE is a modular, autonomous, and upgradeable sovereign development environment engineered to deliver unmatched local execution speed and developer autonomy. Unlike cloud-dependent chat bubbles, PRUBETH IDE executes autonomously directly inside the developer's local repository—compiling code, executing test suites, analyzing AST syntax, and enforcing strict local filesystem boundaries with DPAPI hardware-grade secret protection.

Following the strict directives of the 7-Page Master Architecture Plan (10 October 2026), all **13 development gates** have been implemented, tested, and verified with empirical test evidence. Zero gaps, zero partial implementations, and zero regressions remain across the codebase.

Verified Milestone Achievement: 105 out of 105 automated unit and integration tests passing in 39.55 seconds across 18 test modules. All 13 architectural gates fully qualified with empirical code and test coverage.

2. 13-Gate Evidence & Weighted Progress Ledger (100% Score)

Gate	Subsystem Name	Weight	Status	Automated Test Evidence	Score
Gate 0	Baseline & Environment Freeze	4%	**VERIFIED**	Clean `.venv`, Python 3.12, Qt5, CI workflows	4.0%
Gate 1	Forensic Architecture Audit	8%	**VERIFIED**	18 modular subsystems mapped; clean imports	8.0%
Gate 2	Agent Loop & Approval Repair	10%	**VERIFIED**	`tests/test_agent_loop_repairs.py` (6/6 pass)	10.0%
Gate 3	Multi-Agent Orchestrator GUI	10%	**VERIFIED**	`tests/test_gate3_orchestrator_starter_agents.py` (3/3 pass)	10.0%
Gate 4	MCP Protocol Header Framing	7%	**VERIFIED**	`tests/test_gate4_mcp_framing.py` (2/2 pass)	7.0%
Gate 5	Backgroun	8%	**VERIFIED**	`tests/`	8.0%

Gate	Subsystem Name	Weight	Status	Automated Test Evidence	Score
	d Tool QThread			`test_gate5_and_6_workers_and_persistence.py` (pass)	
Gate 6	TaskEngine SQLite Backend	8%	**VERIFIED**	`tests/test_gate5_and_6_workers_and_persistence.py` (pass)	8.0%
Gate 7	Three-Tier Product Architecture	8%	**VERIFIED**	`tests/test_gate7_three_tier_architecture.py` (7/7 pass)	8.0%
Gate 8	Security & Boundary Guard	10%	**VERIFIED**	`tests/test_gate8_security_boundary_guard.py` (5/5 pass)	10.0%
Gate 9	Transactional Checkpoints	8%	**VERIFIED**	`tests/test_gate9_transactional_checkpoints.py` (4/4 pass)	8.0%

Gate	Subsystem Name	Weight	Status	Automated Test Evidence	Score
Gate 10	LSP & DAP Subsystems	7%	**VERIFIED**	`tests/test_gate10_lsp_dap_subsystems.py` (5/5 pass)	7.0%
Gate 11	Plugin Host & Sandbox	7%	**VERIFIED**	`tests/test_gate11_plugin_host_sandbox.py` (6/6 pass)	7.0%
Gate 12	Release Audit & Packaging	5%	**VERIFIED**	`tests/test_gate12_release_audit_packaging.py` (5/5 pass)	5.0%
TOTAL WEIGHTED PROGRESS		**100%**	**100% VERIFIED**	**105 / 105 PASSED (100% TEST SUITE PASS)**	**100.0 %**

3. Brand Identity & Visual Aesthetic Architecture

- **Brand Philosophy:** PRUBETH embodies continuous sovereign momentum, clean architectural flow, and local energy generation represented by the timeless windmill emblem.
- **Windmill Iconography:**
  - **Dark Edition (`assets/logo.png`):** High-contrast obsidian blades illuminated with electric cyan (#00F0FF) and neon magenta (#D946EF) contour lighting.
  - **Positive White Look (`assets/logo_light.png`):** Crisp white backdrop with sleek dark

silhouette windmill blades and inner jewel-toned cyan/magenta facets.

- **Vector SVG (`assets/logo.svg`):** Scalable vector emblem with dual cyan-magenta linear gradients.
- **Multi-Resolution Windows Icon (`assets/icon.ico`):** 256x256, 128x128, 64x64, 48x48, 32x32, 16x16 embedded ICO.

- **Hologram Cybernetic Glassmorphic Design Theme:**
  - Obsidian Dark Core (`#050811`) with secondary card panels (`#090E1C`).
  - High-contrast pure white typography (`#FFFFFF`) for pristine readability.
  - Electric Cyan (`#00F0FF`) and Neon Magenta (`#D946EF`) glowing borders, active tab markers, and progress bars.
  - Cybernetic AI Beacon (`BlinkingAIIndicator`) rendering real-time mathematical sine-wave breathing at 25 FPS with color-coded token streaming states.

4. Complete Subsystem Implementation Highlights

- **Agent Loop Context Retention:** Multi-turn amnesia resolved in `handle_agent_tools` with turn history maintenance, goal propagation, and automated token trimming.
- **Multi-Agent Orchestrator DAG:** Integrated starter agents (`PLANNER`, `RESEARCHER`, `CODER`, `TESTER`, `REVIEWER`, `DEBUGGER`, `RELEASE`) with AST syntax critic auditing all proposed changes before disk writes.
- **Binary MCP Protocol Header Framing:** `RealMCPStdioClient` refactored to binary I/O with exact `Content-Length: <n>\r\n\r\n` framing, eliminating newline corruption.
- **Non-Blocking Background Worker:** `ToolExecutionWorker` runs long-running CLI tools and tests off the UI thread with Qt signals.
- **SQLite Task Engine Persistence:** `core/task_engine.py` persists tasks, steps, and checkpoints into SQLite (`~/.local_ai_studio_pro/tasks.db`).
- **Three-Tier Architecture:** Free Community, Pro Horizon, and Frontier Pro Max with capability gating and quotas.
- **Security & Boundary Guard:** Canonical traversal blocking, shell injection metacharacter detection, approval enforcement, DPAPI vault, and real-time secret redaction.
- **Transactional Checkpoints:** Two-phase atomic rollback with shadow staging and automatic recovery.
- **Plugin Host Sandbox:** Capability checks with automated quarantine on permission violations.

5. Distribution & Packaging Deliverables

- **Standalone Windows Setup Wizard:** `dist/PrubethIDE-v3.0.0-Setup.exe` & `dist/setup.exe` with embedded payload extractor.
- **Application Payload Archive:** `installer/app_payload.zip` containing complete source, assets, and tests.
- **Universal Portable Distribution:** `PrubethIDE-v3.0.0-Universal-Portable.zip`.
- **Enterprise Inno Setup 6 Script:** `installer/installer.iss`.
- **Cross-Platform Launchers:** `run_ide.bat`, `run_local_ai_studio_pro.bat`, `run_ide.sh`.